Sec: SR STAR - A & APEX, ALL IITS | DT: 18/01/2026
Time: 3 Hrs | GTM – 18 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
C	B	C	C	A	B	A	C	A	A
11	12	13	14	15	16	17	18	19	20
A	B	B	B	B	C	B	B	C	B
21	22	23	24	25					
5	3	4	20	2					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	C	B	C	C	A	A	A	B	B
36	37	38	39	40	41	42	43	44	45
D	A	A	A	A	D	B	A	A	C
46	47	48	49	50					
5	50	92	165	2					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	D	C	A	D	D	C	C	D	B
61	62	63	64	65	66	67	68	69	70
B	D	D	B	C	C	B	C	D	B
71	72	73	74	75					
0	1	45	1	2					

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HINTS
PHYSICS

1. Conceptual

2. At equilibrium $Ah\rho_m g = Ay_0\rho_w g$ ....(1)

After slightly depressed

$$A(y_0 + y)\rho_w g - Ah\rho_m g = Ah\rho_m a$$

$$y\rho_w g = h\rho_m a$$

$$\Rightarrow a = \left(\frac{\rho_w}{\rho_m}\frac{g}{h}\right)y \Rightarrow T = 2\pi\sqrt{\frac{\rho_m h}{\rho_w g}}$$

3. Conceptual

4. $\lambda = \frac{h}{p} = \frac{6.63\times10^{-34}}{2.21\times10^{-30}\times3\times10^{6}} = 10^{-10}m$

5. $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{1.5}{v} = \frac{1}{-20} + \frac{0.5}{5} = \frac{1}{20}$

$\Rightarrow v = 30cm$ $\frac{I}{0} = \frac{\mu_1}{\mu_2}\frac{v}{u} = \frac{1\times30}{-1.5\times20} = -1$

6. $\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mq\Delta V}}$

7. Change on capacitor as a function of time, during glass of change

$$q = q_{\max}\left(1 - e^{-\frac{t}{RC}}\right), I = \frac{q_{\max}}{RC}e^{-\frac{t}{RC}}$$

8. $[A] = [L], [B] = \left[LT^{-1}\right][C] = \left[LT^{-1}\right][0] = [LT]$

9. Conceptual

10. $\tau = C\theta \Rightarrow ML^2T^{-2} = [C][1]$

$$C.S. = \frac{\theta}{I} = \frac{NBA}{C}$$

$V.S. = \frac{NBA}{RC}$ [R also depends on N]

11. $\vec{E}_1 + \vec{E}_2 = 0 \Rightarrow \left|\vec{E}_1\right| = \left|\vec{E}_2\right|$

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$$\Rightarrow \frac{K.2p_1}{\ell^3} = \frac{K.p^2}{\ell^3} \Rightarrow \frac{p_1}{p_2} = \frac{1}{2}$$

12. $\frac{1}{2}mv^2 = ms\Delta\theta + mL$

$$v = \sqrt{2\left(s\Delta\theta + L\right)} = \sqrt{2\left(125\times300 + 2.5\times10^4\right)}$$

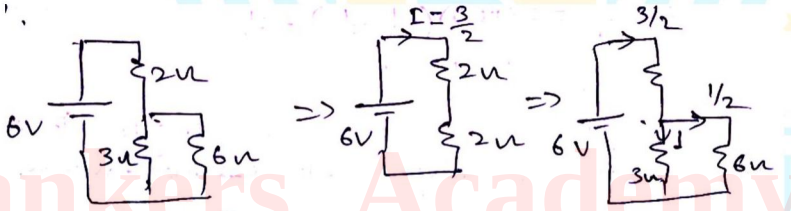
$$v \approx 354 m/s$$

13. lateral shift $d = \frac{t\sin\left(i-r\right)}{\cos r}$

And $\sin t = \mu \sin r$

14. $v = \sqrt{\frac{2gh}{1+\frac{K_2}{R_2}}}, K = radius\ of\ g\ graphs$

15.



16. $\frac{A}{BC} = \frac{\phi}{\rho}\times\frac{\sigma}{\phi}\times\frac{\lambda}{\phi}$

$$= \frac{q}{\phi} = \varepsilon_0$$

17. $p_m = p_\ell + p_m \Rightarrow \frac{-1}{F} = \frac{2}{f\ell} - \frac{1}{fm}$

$$\Rightarrow -\frac{1}{F} = 2\left(1.5-1\right)\left(-\frac{1}{40}+\frac{1}{20}\right) - \frac{1}{-10}$$

$$\Rightarrow -\frac{1}{F} = \frac{1}{40}+\frac{1}{10} = \frac{5}{40}$$

$\Rightarrow F = -\theta cm$ concave mirror of focal length 8 cm

18. $\hat{K} = \frac{3i+4i}{5}, \hat{E} = \frac{4i-3j}{5}$

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$$\hat{B}=\hat{K}\times\hat{E}=\frac{1}{25}\left(3\hat{i}+4\hat{j}\right)\times\left(4i-3j\right)$$

$$\hat{B}=\frac{1}{25}\left[R-9\hat{K}+16\left(-\hat{K}\right)\right]=-\hat{K}$$

$$B_0=\frac{E_0}{C}=\frac{47}{3\times10^8}$$

19. Area under curve

20. Conceptual

21. $$\frac{\frac{1}{4\pi t_0}\frac{q_1q_2}{r^2}}{G\frac{m_1m_2}{r^2}}=\frac{9\times10^9\times4.8\times10^{-19}\times9.6\times10^{-10}}{19.2\times10^{-27}\times9\times10^{-27}\times6.6\times10^{-11}}$$

$$=3.6\times10^{44}$$

22. $$\vec{r}=5t^2\hat{i}-5t\hat{j}$$

$$\vec{v}-\frac{d\vec{r}}{dt}=10t\hat{i}-5\hat{j}$$

$$\cos\theta=\frac{\vec{v}.\vec{r}}{|v||r|}=\frac{75}{5\sqrt{5}.5\sqrt{2}}=\frac{3}{\sqrt{10}}$$

23. $$\Delta T=T_2-T_1\Rightarrow T_2=4T+T_1=546$$

$$\Rightarrow T_1V_1^{V-1}=T_2V_2^{V-1}\Rightarrow V_2=\frac{V_1}{4}$$

24. $$w=\int_0^2 x\left(10-x\right)dx+\int_0^2 y^2dy$$

$$=\left(5x^2-\frac{x^3}{3}\right)_0^2+\left(\frac{y^3}{3}\right)_0^2$$

$$=\left(20-\frac{8}{3}\right)+\frac{8}{3}=20$$

25. $$Q=\frac{\Delta\phi}{R}$$

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CHEMISTRY

26. Pt – 6th period, 10th group
Ds – 7th period, 10th group
Rb – 5th period, 1st group
Os – 6th period, 8th group

27. $\bar{\nu} = R_H\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$

For Balmer transition $3 \rightarrow 2$

$\bar{\nu} = R_H\left(\frac{1}{2^2}-\frac{1}{3^2}\right)$

$=\frac{5R_H}{36}$

$\lambda=\frac{36}{5R_H}$

28. HNO_2 disproportionate to give HNO_3 and NO

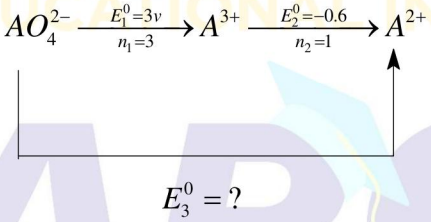
29. $\Delta T_f = i \times T_f \times m$

$3.72 = i \times 1.86 \times 1$

$i = \frac{3.72}{1.86} = 2$

Complex is $\left[Pt(H_2O)_3Cl_3\right]Cl.3H_2O$

30.



$\Delta G_3^o = \Delta G_1^o + \Delta G_2^o$

$-n_3E_3^0 = -n_1E_1^o - n_2E_2^o$

$4E_3^o = 3\times3 - 1\times0.6$

$E_3^0 = \frac{9-0.6}{4} = \frac{8.4}{4} = 2.1$

31. Conceptual

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32. Fructose doesnot contain $-CHO$ Group but reduces Tollens reagent.

33. No. of millimoles of N_2O taken initially= $\frac{x\times10^{-3}}{44}$.

No. of millimoles of N_2O removed $=\frac{6\times10^{20}}{6\times10^{23}}$

$=10^{-3}=1$ millimole.

$$\frac{x\times10^{-3}}{44}=1.5\times10^{-3}+10^{-3}$$

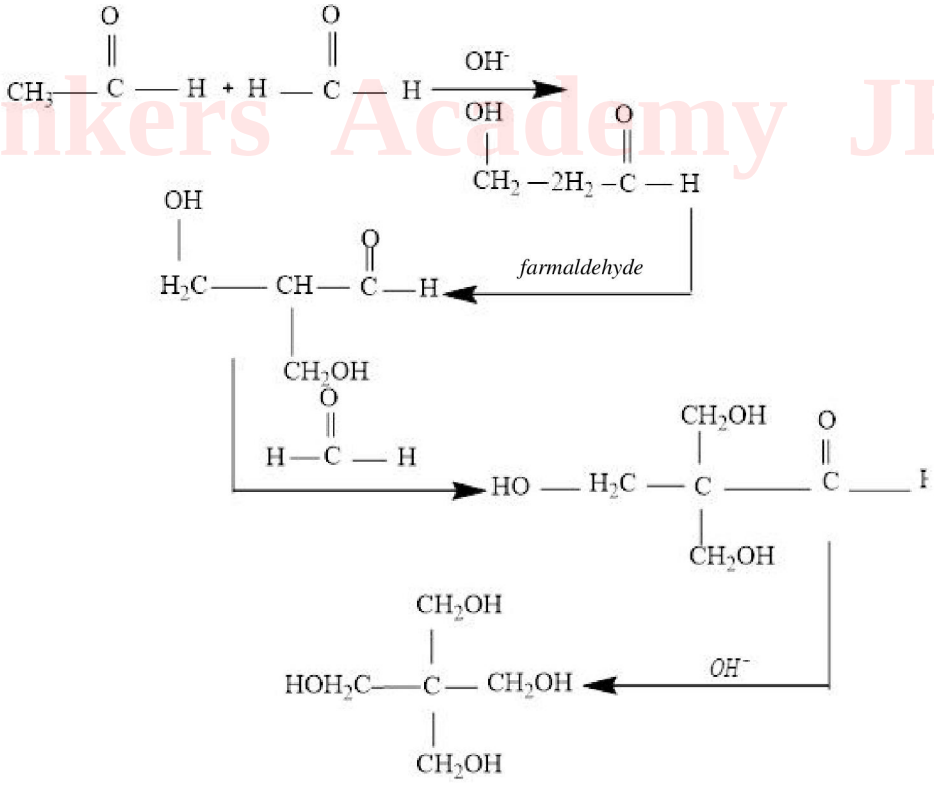
$$x=2.5\times44=110$$

34. $\underset{270K}{Ice}\rightarrow\underset{273K}{Ice}\rightleftharpoons\underset{273K}{Water}\rightarrow\underset{373K}{Water}\rightleftharpoons\underset{373K}{WaterVapour}\rightarrow\underset{390K}{Water\ Vapour}$

$$\Delta S=\int_{270}^{273}\frac{C_{pm}dT}{T}+\frac{\Delta H_{fusion}}{273}+\int_{273K}^{373K}\frac{C_{pm}dT}{T}+\frac{\Delta H_{vapourisation}}{373}+\int_{373K}^{390K}\frac{C_P.dT}{T}$$

36. $\left[Co\left(NH_3\right)_3Cl_3\right]$ shows fac-mer isomerism. where as $\left[Pt\left(NH_3\right)_3Cl\right]Cl$, $\left[Zn\left(NH_3\right)_2Cl_2\right], k_3\left[Fe\left(C_2O_4\right)_3\right]$ do no show geometrical isomerism.

37.



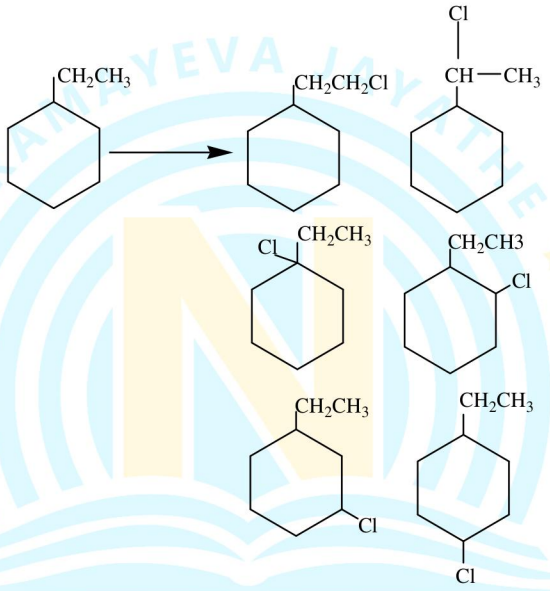
38. Carbo cations stability

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$$CH_3-\overset{+}{C}-CH_3 \quad (CH_3 \text{ below } C)$$

$$CH_3\overset{+}{C}H_2 < CH_3-\overset{+}{C}H-CH_3 <$$

39.



40. no.of moles of phenol $= \frac{9.4}{94} = 0.1$

$$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 8HBr$$

$1 mole \quad 3 moles$

1 mole phenol requires 3 mole of Br_2

0.1 mole phenol requires =0.3 mol

Weight of Br_2 in g = $0.3\times160 = 48g$

42. Conceptual

43. $Q_{I.P} > K_{sp}$

$\Rightarrow A(OH)_3$

$\left[A^{3+}\right]\left[OH^-\right]^3 > 27\times10^{-18}$

$\left[OH^-\right] > 3\times10^{-6}$

$B(OH)_2$

$\left[B^{2+}\right]\left[OH^-\right]^2 > 9\times10^{-10}$

$\left[OH^-\right] > 3\times10^{-5}$

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$\therefore A(OH)_3$ precipitates first

44.

$X: sp^2 - sp^3$ $Y: sp^3 - sp^3$	X: Trigonal planar to tetrahedral Y: Trigonal pyramidal to tetrahedral

45. Primary and secondary amines react with Hinsberg Reagent

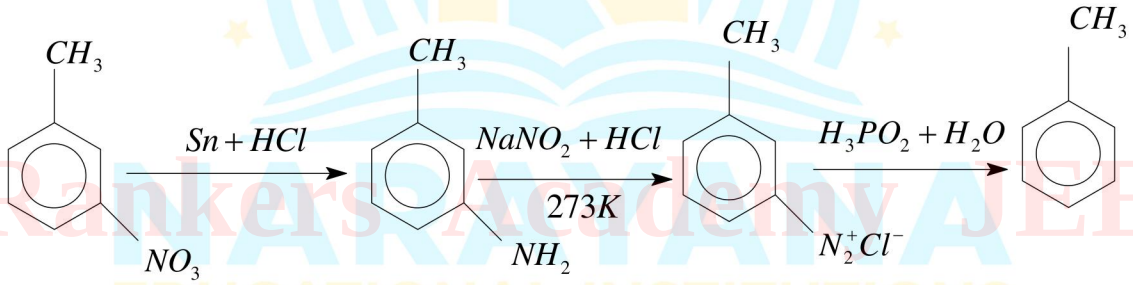
47. $M.eq.$ of $K_2Cr_2O_7 = M.eq.$ of FeC_2O_4

$$FeC_2O_4 + Cr_2O_7^{-2} \rightarrow Fe^{3+} + CO_2 + Cr^{3+}$$

$$V \times 0.02 \times 6 = \frac{0.288}{144} \times 3 \times 1000$$

$$V = 50mL.$$

48.



49. $2.303 \times 10^{-2} = \frac{2.303}{60} \log \frac{1.2}{1.2 - p}$

$$60 \times 10^{-2} = \log \frac{1.2}{1.2 - p}$$

$$\frac{1.2}{1.2 - p} = 10^{0.6}$$

$$\frac{1.2}{1.2 - p} = 4$$

$$4.8 - 4p = 1.2$$

$$4p = 4.8 - 1.2 = 3.6$$

$$p = \frac{3.6}{4} = 0.9$$

$$\text{Total pressure} = 1.2 + \frac{p}{2}$$

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$$=1.2+\frac{0.9}{2}=1.65atm$$

$$x=165\times10^{-2}atm$$

50.

$$\underset{\text{intially}\\ \text{At equilibrium}}{\overset{}{\underset{1}{CO}(g)+\underset{1}{NO_2}(g)\rightleftharpoons\underset{1}{CO_2}(g)+\underset{1}{NO}(g)}}$$
$$\underset{0.8}{}\qquad\underset{0.8}{}\qquad\underset{1.2}{}\qquad\underset{1.2}{}$$

$$\underset{1.2}{Ba(OH)_2}+\underset{1.2}{CO_2}\rightarrow\underset{1.2}{BaCO_3}+\underset{\text{moles}}{H_2O}$$

$$\therefore K_c=\frac{1.2\times1.2}{0.8\times0.8}=2.25$$

MATHEMATICS

51. Put In $x=t\Rightarrow x=e^t\Rightarrow dx=e^t dt$

$$I=\int_{-\infty}^{\infty}f\left(e^t+e^{-t}\right)\frac{t}{e^t}dt=\int_{-\infty}^{\infty}f\left(e^t+e^{-t}\right)tdt=0$$

(as the function is odd)

Alternatively-1 : put $x=\tan\theta$;

$$\int_0^{\pi/2}f\left(\tan\theta+\frac{1}{\tan\theta}\right)\frac{\ln\tan\theta}{\tan\theta}.\sec^2\theta d\theta$$

$$=\int_0^{\pi/2}f\left(\tan\theta+\frac{1}{\tan\theta}\right)\frac{\ln\tan\theta}{\sin\theta\cos\theta}d\theta$$

Alternative-2: Put $x=1/t\Rightarrow I=-I\Rightarrow 2I=0\Rightarrow I0$

52. Put $x-4=\frac{1}{t}\Rightarrow dx=\frac{1}{t^2}dt$

$$I=f\frac{-\frac{1}{t^2}dt}{\left(\frac{1}{t}\right)^{3/2}\left(-1-\frac{1}{t}\right)^{3/2}}=-\int\frac{dt}{\sqrt{-t-1}}=\frac{-2\sqrt{-t-1}}{-1}+c$$

$$=2\sqrt{\frac{1}{4-x}+1}+c=2\sqrt{\frac{x-3}{4-x}}+c$$

53. We have $f(x)=\begin{cases}3\left(1+|\tan x|\right)^{\frac{\alpha}{|\tan x|}};\ \frac{-1}{2}<x<0\\ \beta;x=0\\ 3\left(1+\left|\frac{\sin x}{3}\right|\right)^{\frac{6}{|\sin x|}};0<x<\frac{2}{3}\end{cases}$

Now, RHL = $f\left(0^+\right)=\lim_{x\to0^+}f(x)$

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$$= = \lim_{x\to 0^+} 3\left(1+\frac{|\sin x|}{3}\right)^{\frac{6}{|\sin x|}} = 3e^2$$

Similarly, LHL $= f\left(0^-\right) = \lim_{x\to 0^-} f\left(x\right) = 3e^{\alpha}$

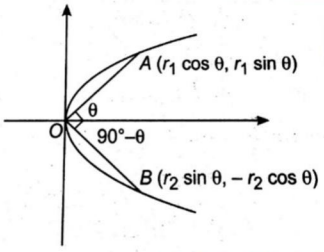
So, for the function $f\left(x\right)$ to be continuous at $x=0$, we must have

$\Rightarrow \quad f\left(0^+\right) = f\left(0^-\right) = f\left(0\right)$

$3e^{\alpha} = 3e^2 = \beta$

So, $\alpha = 2, \beta = 3e^2$

54.



$$r_1 = \frac{4a\cos\theta}{\sin^2\theta}$$

Similarly, $r_2 = \frac{-4a\sin\theta}{\cos^2\theta}$

Put in $\frac{r_1^{4/3} r_2^{4/3}}{r_1^{2/3} + r_2^{2/3}}$.

55. $\frac{dy}{dx} = \frac{\left(\sin 2x - \cos^2 x\right) y}{e^{-x}\left(y+1\right)}$

$$\int \frac{y+1}{y} dy = \int e^x \left(-\cos^2 x + \sin 2x\right) dx$$

$$y + \ln y = e^x\left(-\cos^2 x\right) + C$$

If $x=0, y=1$

$\Rightarrow C = 2$

56. $\sin^{-1} x > 0$ and $\cos^{-1} x > 0$

Now, $0 < \sin^{-1} x \le \frac{\pi}{2}$

$\Rightarrow \quad 0 < x \le 1$

And $0 < \cos^{-1} x \le \pi$

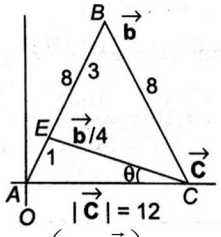
$\Rightarrow \quad -1 \le x < 1$

Combining (i) and (ii)

$0 < x < 1$

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57. Given $\left|\vec{b}\right| = \left|\vec{b}-\vec{c}\right| = 8$ and $\left|\vec{c}\right| = 12$



$$\cos\theta = \frac{\vec{c}.\left(\vec{c}-\frac{\vec{b}}{4}\right)}{\left|\vec{c}\right|\left|\vec{c}-\frac{\vec{b}}{4}\right|} = \frac{\vec{c^2}-\frac{\vec{c}.\vec{b}}{4}}{12\left|\vec{c}-\frac{\vec{b}}{4}\right|}$$

Now proceed.

58. $S_{16} = \frac{16}{2}[2a-15d] = 120$

$\Rightarrow 2a+15d = 15 \Rightarrow \frac{2}{15}a+d = 1$

59. $\begin{vmatrix} a-2 & 7-9 & -2-13 \\ 1 & 2 & 3 \\ -1 & 2 & -3 \end{vmatrix} = 0$

$(a-2)(-12)+2(-3+3)-15(2+2) = 0$

$\Rightarrow -12a+24-60 = 0 \Rightarrow a = -3$

60. $\left|\omega z-1-\omega^2\right| = a \Rightarrow |z+1| = a$

$\Rightarrow |z-3+4| = a \Rightarrow |z-3|+4 \geq a$

$\Rightarrow 0 \leq a \leq 8$

61. Given $R = [(a,b),(a,c),(b,b)]$

$(a,a),(c,c),(d,d),(e,e)$ for reflexive

$(b,a),(c,a) \rightarrow$ for symmetric

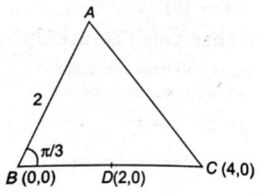
$(c,b),(b,c) \rightarrow$ for transitive

$\therefore$ 8 minimum ordered pairs

62. When A has B or C to his right we have AB or AC
When B has C or D to his right we have BC or BD
Thus $\Rightarrow$ We must have ABC or ABC or AC and BD
For A,B,C, D,E,F on a circle number of ways = 3!=6
For A,B,D,C,E,F on a circle number of ways = 3!=6
For AC,BD, E,F the number of ways = 3!=6
Total=18

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63.



Coordinates of A and C are respectively $(2\cos\ 60^0,\ 2\sin 60^0)$ and $(4,0)$

i.e. $A \equiv \left(1,\sqrt{3}\right)$ and $C \equiv (4,0)$ therefore centroid is

$$\left(\frac{1+4}{3},\frac{\sqrt{3}}{3}\right) \equiv \left(\frac{5}{3},\frac{1}{\sqrt{3}}\right)$$

64. For existence of $\sec^{-1}(\sqrt{2x^2-x^4})$ or $\cos ec^{-1}(\sqrt{2x^2-x^4})$

$$2x^2 - x^4 \geq 1$$

i.e., $$\left(x^2-1\right)^2 \leq 0$$

Hence, $x^2 = 1 \Rightarrow x = \pm 1$, if $x = 1, a + \frac{\pi}{2} = 0$;

If $x = -1, -a + \frac{\pi}{2} = 0 \Rightarrow a = \pm\frac{\pi}{2}$

65. $\sin x + \sin y + \sin z + \sin w = -4$

$\Rightarrow \sin x = \sin y = \sin z = \sin w = -1$ (only)

So $\sin^{400} x + \sin^{300} y + \sin^{200} x + \sin^{100} w$

$= (-1)^{400} + (-1)^{300} + (-1)^{200} + (-1)^{100} = 4$

66.

$\mu = 5.5$

$\sum x_i = 55$

Correct sum

$\sum x_i = 55 - 4 - 5 + 6 + 8 = 60$

Correct mean = 6

Correct $\sum x_1^2 = 371 - 16 - 25 + 36 + 64 = 471 - 41$

Variance= $\frac{430}{10} - 36 = 7$

67. $3G = 2S + H$

$$\vec{p} + \vec{q} + \vec{r} = \frac{\vec{p} + \vec{q} + \vec{r}}{2} + \alpha\vec{p} + \beta\vec{q} + \gamma\vec{r}$$

Computing $\alpha = \frac{1}{2}, \beta = \frac{1}{2}, \gamma = \frac{1}{2}$

$$\alpha + 2\beta + 5\gamma = \frac{1}{2} + 1 + \frac{5}{2}$$

$= 4$

68. $A^2 = 2A - I$

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$A^3 = 2A^2 - IA$

$= 2(2A - I) - A(A^2 = 2A - I)$

$A^3 = 3A - 2I$

$A^4 = 3A^2 - 2A$

$= 3(2A - I) - 2A(A^2 - 2A - I)$

$A^4 = 4A - 3I$

$A^5 = 5A - 4I$

.....

.....

$A^n = nA - (n-1)I$

69. $\begin{vmatrix} k & 2 & -1 \\ (k-1) & k & 1 \\ 1 & (k-1) & k \end{vmatrix} \neq 0$

$\Rightarrow$ k has infinite solutions

70. x denotes the number on red die

Y denotes the number on blue die.

Then $x + y \leq 6$ (as the number on green has to be less than or equal to 6) but $x \geq 1$ and $y \geq 1$, hence

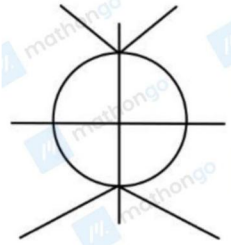
$x + y \leq 4$ (using beggar)

$x + y + t = 4$

$\Rightarrow {}^6C_2 = 15 = n(A)$

$\Rightarrow P = \frac{15}{216} = \frac{5}{72} = \frac{a}{b} \Rightarrow a + b = 77$

71.



So area is zero

72. $(9 + 4\sqrt{5})^n = 3 + f$

Let $(9 - 4\sqrt{5})^n = f'$

$I + f + f' = (9 + 4\sqrt{5})^n + (9 - 4\sqrt{5})^n$ = even integer

$0 < f + f' < 2 \Rightarrow f + f' = 1$

$\therefore$ I is an odd integer.

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73. $\frac{x^2}{5}-\frac{y^2}{5\cos^2\alpha}=1$

$\frac{x^2}{25\cos^2\alpha}+\frac{y^2}{25}=1$

Given $1+\frac{5\cos^2\alpha}{5}=3\left(1-\frac{\cos^2\alpha}{2\cos^2\alpha}\right)$

$1+\cos^2\alpha=3-3\cos^2\alpha$

$4\cos^2\alpha=2\Rightarrow\cos\alpha=\frac{1}{\sqrt{2}}\Rightarrow\alpha=45^0$

74. $\frac{x^2}{(x+1)^2}-\left(1+\frac{1}{1+x}\right)^2$

$3-x^2=1-\frac{2}{1+x}+\frac{1}{(1+x)^2}$

$2=x^2+\frac{1}{(1+x)^2}-2\left\{\frac{1+x-x}{1+x}\right\}$

$2=x^2+\frac{1}{(1+x)^2}+\frac{2x}{1+x}-2$

75. Let discriminant of $10px^2-qx+r$, $px^2-qx-5r$, $5px^2-qx-r$ be D_1, D_2 and D_3 respectively.

$\therefore\ D_1+D_2+D_3=q^2-40pr+q^2+20pr$

$+q^2+20p=3q^2>0$

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